F70LA Life Insurance Mathematics A (2025/26)
Tutorial 2

1. Express the following probabilities using international actuarial notation.

   (i) The probability that a life currently aged 30 will die before exact age 60.

   (ii) The probability that a life currently aged 50 will survive to at least age 80.

   (iii) The probability that a life currently aged 40 will die between exact ages 55 and 65.

   (iv) The probability that a life currently aged 30 will die either between exact ages 40 and 45 or between exact ages 50 and 55.

2. T_0 is the random variable representing the complete future lifetime of a newly born child. T_x is the corresponding random variable for a life currently aged x. You are given that the cumulative distribution function of T_x is

$$F_x(t) = 1 - e^{-\alpha t - \beta(2xt+t^2)},$$

   where α and β are constants. Given that $\alpha = 0.002$ and $\beta = 2.3 \times 10^{-5}$, calculate:

   (i) ${}_{40}p_0$, (ii) ${}_{20}p_{60}$, (iii) ${}_{5}|{}_{5}q_{50}$.

3. Suppose that, for integer ages $x \in \{80, 81, 82, 83, 84\}$, it is known that

$$p_x = \frac{89 - x}{10}.$$

   Calculate the values of

   (a) q_{80}, (b) ${}_2p_{80}$, (c) ${}_5p_{80}$, (d) ${}_3q_{82}$, (e) ${}_{2}|{}_{3}q_{80}$.

4. Given that

$$\mu_x = \begin{cases} 0.001 & \text{for } x \leq 40 \\ 0.001 + 0.01(x - 40) & \text{for } x > 40, \end{cases}$$

   calculate:

   (i) $S_0(50)$, (ii) ${}_{12}q_{35}$.

5. Show that

$$\frac{\partial}{\partial x}{}_tp_x = {}_tp_x \left(\mu_x - \mu_{x+t}\right).$$

6. Given that
$$\mu_x = \begin{cases} 0.03 & \text{if } x \leq 70 \\ 0.03 + 0.003(x - 70) & \text{if } x > 70, \end{cases}$$
evaluate:

(i) ${}_{10}p_{50}$, (ii) ${}_{10}p_{70}$, (iii) ${}_{7}q_{70}$, (iv) ${}_{8}|{}_{7}q_{66}$.

7. Fix $x \geq 0$. Show that if ${}_tp_x\mu_{x+t}$ is strictly increasing on the interval $0 \leq t \leq 1$ then $q_x > \mu_x$.

8. Consider a simple exponential mortality model with $\mu_x = \lambda$ for all $x \geq 0$.

(i) Write down the limiting age of this model.

(ii) Derive an expression for $\mathring{e}_x$ in terms of λ.

(iii) Derive an expression for e_x in terms of λ.

(iv) Suppose you wish to model mortality for ages $x \geq 75$ only using the simple exponential mortality model. What would you choose for the parameter λ, if data shows that the expected future lifetime from age 75 is 10 years?

9. Given that the probability of death within one year is 0.02 at all ages, evaluate

(i) ${}_{55}p_{30}$, (ii) ${}_{25}p_{50}$, (iii) ${}_{15}q_{30}$, (iv) ${}_{10}|{}_{3}q_{40}$.

10. Suppose that $\mu_x = 0.02x$ for all $x \geq 0$. Derive an expression for the density function of the random variable T_x.

Numerical solutions: 1(i) ${}_{30}q_{30}$; 1(ii) ${}_{30}p_{50}$; 1(iii) ${}_{15}|{}_{10}q_{40}$; 1(iv) ${}_{10}|{}_{5}q_{30} + {}_{20}|{}_{5}q_{30}$. Q2(i) 0.8898; 2(ii) 0.9009; 2(iii) 0.02246. Q3(i) 0.1; 3(ii) 0.72; 3(iii) 0.1512; 3(iv) 0.79; 3(v) 0.5688. Q4(i) 0.57695; 4(ii) 0.22663. Q6(i) 0.74082; 6(ii) 0.63763; 6(iii) 0.24686; 6(d) 0.23618. Q8(i) ∞; Q8(iv) 0.1; Q9(i) 0.3292; 9(ii) 0.6035; 9(iii) 0.2614; 9(iv) 0.0481.